MATHUSAN MANIVANNAN

Data Science Undergraduate | Intern – Data Science

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PROFESSIONAL SUMMARY

Data Science undergraduate, driven by a passion for building real-world solutions through technology. A builder and entrepreneur at heart, with a track record of turning ideas into live, working products. Committed to growing as both a data scientist and a businessman - creating things that genuinely matter.

EDUCATION

BSc (Hons) in Information Technology, Specialization in Data Science 2024 - Present

Sri Lanka Institute of Information Technology (SLIIT), Northern Uni, Jaffna.

PROJECTS

Student Stress Level Predictor Oct 2025

scikit-learn, Random Forest, SVM, KNN

- Analysed 1,100 student records across 12 features covering anxiety levels, sleep quality, academic pressure, and career concerns to understand patterns in student stress
- Trained and compared 5 classifiers Random Forest, SVM, KNN, Logistic Regression, and Decision Tree, with Random Forest achieving the highest accuracy at 88.75%
- Applied data preprocessing and class balancing, with consideration of ethical risks including model bias and the need for human oversight in mental health prediction contexts.

StaySync AI - Hostel & Restaurant Management System Jan 2026 - Present

Python, React, Ollama (LLM), XGBoost, scikit-learn, PuLP | SLIIT Group Project

- Developed a room recommendation engine using Random Forest Regression with a two-pass ranking system, achieving an R^2 of 0.871, MAE of 0.083, and RMSE of 0.112.
- Built a food and restaurant matching system using Cosine Similarity, reaching 90% ranking accuracy across meal and dining recommendations.
- Designed a budget-aware meal planner that factors in location using Haversine distance calculations and enforces nutritional and cost constraints through PuLP optimization.
- Built the React frontend across student, admin, and restaurant user roles.

Full-Stack POS & Student Order System May 2026

Python, React, PostgreSQL | Client Project

- Designed and delivered a production-ready Point-of-Sale and student order management system for a real client, built and deployed as part of a peer startup
- Managed the full software development lifecycle from requirement gathering through to deployment on Netlify and Railway, with the client signing off satisfied on final delivery
- Worked in an agile-style team environment handling client communication, task coordination, and deployment, taking the project from first meeting to live product

Fake Job Posting Detector - ML Classification Project Jun 2026

Python, scikit-learn, TF-IDF, SMOTE, Random Forest, Logistic Regression

- Engineered a fraud detection pipeline on 17,880 job postings using TF-IDF vectorization and feature engineering, targeting a 4.8% fraudulent class imbalance.
- Trained and benchmarked 3 classifiers (Logistic Regression, Random Forest, Random Forest + SMOTE), evaluating on F1-score and ROC-AUC to account for class imbalance
- Achieved 99.4% ROC-AUC, 100% precision, and 70.5% recall using Random Forest with SMOTE oversampling.
- Engineered domain-driven features such as missing company profile, salary range, and logo presence to improve model interpretability and reflect real-world fraud signals.

ADDITIONAL INFORMATION

- Languages: English (professional), Tamil (native)
- Available for internships, part-time roles, and collaborative AI/ML projects

TECHNICAL SKILLS

Languages: Python, SQL, Java, JavaScript

ML & AI: scikit-learn, Random Forest, XGBoost, SVM, KNN, Logistic Regression

Data & Analytics: Pandas, NumPy, Matplotlib, Exploratory Data Analysis, Feature Engineering

Databases: MySQL, PostgreSQL, MongoDB

Statistics: Probability, Hypothesis Testing, Confidence Intervals, Statistical Distributions

Web & Dev Tools: React.js, REST APIs, Git, GitHub, Postman, Netlify, Railway

REFERENCES

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